

OTA update server

BMS fleet

Data plane (Tofino programmable switch)

Ingress parser
(Eth/IP/TCP/MQTT/OTA)

Rules R1, R2, R4, R5, R6
(MAT + per-BMS registers,
incl. R6 version high-water)

Policy engine
PASS / HOLD / DROP

Session override table
(5-tuple-keyed; source-IP aggregation optional)

Deparser + digest
(6 types, ≤ 48 B each)

Control plane (server CPU)

time tick + state reset

OTA-Shield controller
(Python + BF Runtime)

RAT-aware arbiter
Gate A (R1) + Gate B (R6)
HOLD $\rightarrow$ PASS or DROP

Rollout-Authorization Table
(policy manifest)

*install override
5 s fail-closed TTL*

gRPC digest stream

decision log
(ground-truth record)